

CHM111: General Chemistry I: Practice Exam

1. 6.46×10^{-7} moles
2. 1.3×10^{26} g
3. 3.902 mL of aluminum iodide
4. CsI
5. 19%
6. $3\text{C} + 4\text{Al}(\text{OH})_3 \rightarrow 6\text{H}_2\text{O} + 3\text{CO}_2 + 4\text{Al}$
7. 2.189 g
8. 0 M
9. $4\text{O}_2 + 3\text{H}_2 + \text{N}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{HNO}_3$
10. 34.1 g/mol
11. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
12. 1.40×10^2 g
13. 7.3×10^{-57} moles
14. 0.969 M
15. 18.79 g
16. 22.95 g
17. $2\text{H}_2\text{O}(l) + 2\text{K}(s) \rightarrow 2\text{KOH}(aq) + \text{H}_2(g)$
18. $0.26 \frac{\text{mol}}{\text{L}}$
19. 16.84 g
20. 353 kg
21. 62.02 g/mol
22. 9.5236 mol
23. 1.37×10^1 mol
24. $\text{S}_8 + 12\text{O}_2 + 8\text{H}_2\text{O} \rightarrow 8\text{H}_2\text{SO}_4$
25. $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
26. 0.4580 moles
27. $x = 4$
28. 8.93 g
29. H_3O^+ , O^- , and H_2O
30. $2\text{KOH} + \text{H}_2\text{CrO}_4 \rightarrow 2\text{H}_2\text{O} + \text{K}_2\text{CrO}_4$
31. $\text{H}_3\text{O}_5\text{Mn}$
32. 13.0%
33. $1.97 \frac{\text{mol}}{\text{L}}$
34. 9.9%
35. Oxygen
36. 679 kg
37. $\text{Sr}(s) + \text{Cu}^{2+}(aq) \rightarrow \text{Sr}^{2+}(aq) + \text{Cu}(s)$
38. neither oxidized nor reduced
39. 154.3 mL
40. 0.167 M
41. $0.361 \frac{\text{mol}}{\text{L}}$
42. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
43. molecular

44. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
45. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
46. 11.81 g
47. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
48. 31.4 mL
49. $6\text{H}_2\text{O} + 5\text{O}_2 + \text{P}_4 \rightarrow 4\text{H}_3\text{PO}_4$
50. $8.18 \times 10^{-4} \text{ L}$
51. 7
52. $\text{Fe}^{3+}(\text{aq}) + 3\text{F}^{-}(\text{aq}) \rightarrow \text{FeF}_3(\text{s})$
53. $3.23 \times 10^1 \text{ g}$
54. $4\text{O}_2 + 3\text{H}_2 + \text{N}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{HNO}_3$
55. 69.8%
56. C_2H_2
57. Oxidizing Agent: Br_2 Reducing Agent: Na
58. Decomposition
59. ionic
60. 0.423 mol
61. 17.3 g
62. $4.32 \times 10^1 \text{ g}$
63. $0.0324 \frac{\text{mol}}{\text{L}}$
64. $3\text{C} + 4\text{Al}(\text{OH})_3 \rightarrow 6\text{H}_2\text{O} + 3\text{CO}_2 + 4\text{Al}$
65. $2\text{HNO}_2 + \text{Co}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{Co}(\text{NO}_2)_2$
66. 8.960 g
67. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
68. 94.0%
69. 0.0403 g
70. $6\text{H}_2\text{O} + 5\text{O}_2 + \text{P}_4 \rightarrow 4\text{H}_3\text{PO}_4$
71. Oxidized: Zn Reduced: Sb
72. 52%
73. oxidized
74. $2.07 \times 10^1 \text{ g}$
75. 17.2 mL
76. 225.0 g
77. 1.038 moles
78. 4.882×10^{19} sulfur atoms
79. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
80. $3.94 \times 10^{27} \text{ g}$
81. 0.3108 mol
82. $\text{H}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\ell) \rightarrow \text{SO}_4^{2-}(\text{aq}) + 2\text{H}_3\text{O}^{+}(\text{aq})$
83. 36.46 g/mol
84. A_3B_2
85. 77%
86. $2.65 \times 10^1 \text{ g}$
87. A_2B_4
88. $\text{Zn}(\text{OH})_2 + 2\text{HNO}_3 \rightarrow 2\text{H}_2\text{O} + \text{Zn}(\text{NO}_3)_2$

89. NH_4Cl
90. $0.59 \frac{\text{mol}}{\text{L}}$
91. 37%
92. 18.9 g
93. 53.1 mL
94. $\text{Fe}_2\text{O}_3 + 3 \text{CO} \rightarrow 2 \text{Fe} + 3 \text{CO}_2$
95. $\text{S}_8 + 12 \text{O}_2 + 8 \text{H}_2\text{O} \rightarrow 8 \text{H}_2\text{SO}_4$
96. 4.62 moles
97. 0.167 moles
98. 8.0×10^{21} oxygen atoms
99. $x = 5$
100. SbI_3
101. $1.093 \times 10^3 \text{ kg}$
102. 6.9×10^{-7} moles
103. 1.6 moles
104. 9.15×10^{-33} moles
105. $2 \text{H}_2\text{O}(\ell) + \text{Ba}(s) \rightarrow \text{Ba}(\text{OH})_2(aq) + \text{H}_2(g)$
106. CF_4
107. 4.8%
108. 3.68 mM
109. $\text{Pb}(\text{OH})_2 + 2 \text{HNO}_3 \rightarrow 2 \text{H}_2\text{O} + \text{Pb}(\text{NO}_3)_2$
110. 2.41 mol
111. molecular, strong acid
112. $x = 8$
113. Double-Displacement
114. No reaction occurs.
115. $\text{TiO}_2 + 2 \text{Cl}_2 + \text{C} + 2 \text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2 \text{MgCl}_2$
116. $6 \text{MnCO}_3 + 8 \text{Al} + 3 \text{O}_2 \rightarrow 4 \text{Al}_2\text{O}_3 + 6 \text{CO}_2 + 6 \text{Mn}$
117. 103.1 mL
118. Oxidizing Agent: CO_2 Reducing Agent: Be
119. 39%
120. 0.1658 M
121. 20.22 g
122. 18.08 g
123. 0.085 mol
124. $2.58 \times 10^{-6} \text{ g}$
125. Oxidizing Agent: O_2 Reducing Agent: Ca
126. $4.74 \times 10^{-9} \text{ g}$
127. $7.76 \times 10^{-1} \text{ mol}$
128. 59.1%
129. $3.90 \times 10^0 \text{ mol}$
130. 0.464 g
131. $\text{HNO}_3(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{NO}_3^-(aq) + \text{H}_3\text{O}^+(aq)$
132. 12.5 g
133. molecular, weak base

134. $152.04 \frac{\text{g}}{\text{mol}}$
135. 16.1 g
136. 8.369 g
137. 6.86×10^{-3} moles
138. $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
139. 36
140. $143.60 \frac{\text{g}}{\text{mol}}$
141. $\text{Mg}_3\text{P}_2\text{O}_8$
142. 37.1%
143. No reaction occurs.
144. 18.7 mM
145. $2\text{CH}_3\text{OH} + 3\text{O}_2 \rightarrow 4\text{H}_2\text{O} + 2\text{CO}_2$
146. $2\text{H}_2\text{O}(l) + 2\text{Rb}(s) \rightarrow 2\text{RbOH}(aq) + \text{H}_2(g)$
147. 5.26×10^{10} g
148. $4\text{O}_2 + 3\text{H}_2 + \text{N}_2 \rightarrow 2\text{H}_2\text{O} + 2\text{HNO}_3$
149. 0.318 mol
150. $\text{Pb}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow 2\text{H}_2\text{O} + \text{PbSO}_4$
151. A_2B_3
152. 79.6%
153. 20
154. 200.62 mL
155. $\text{S}_8 + 12\text{O}_2 + 8\text{H}_2\text{O} \rightarrow 8\text{H}_2\text{SO}_4$
156. Nitrogen
157. 80%
158. 5.7×10^{49} g
159. $\text{HClO}_4(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{ClO}_4^-(aq) + \text{H}_3\text{O}^+(aq)$
160. $158.1 \frac{\text{g}}{\text{mol}}$
161. $2\text{Cr}^{3+}(aq) + 3\text{S}^{2-}(aq) \rightarrow \text{Cr}_2\text{S}_3(s)$
162. 7.8×10^{20} iodine atoms
163. Oxidized: Na Reduced: Mn
164. 19%
165. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
166. 55 mM
167. 0.11 M
168. 0.1599 g
169. molecular
170. $\text{H}_2\text{CO}_3 + \text{Mg}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{MgCO}_3$
171. 153.5 g
172. Oxygen
173. 5.1×10^{27} g
174. 54.72 mL
175. 0.247 M
176. 6.2×10^2 moles
177. +2

178. $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
179. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
180. 0.2570L
181. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
182. $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
183. 9.810g
184. Ag_3PO_4 and H_2O
185. $3\text{HClO} + \text{Bi}(\text{OH})_3 \rightarrow 3\text{H}_2\text{O} + \text{Bi}(\text{ClO})_3$
186. 20.67 g
187. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
188. 0.0593 moles
189. 54%
190. molecular
191. 8.82 moles
192. $6\text{H}_2\text{O} + \text{C}_3\text{H}_8 \rightarrow 10\text{H}_2 + 3\text{CO}_2$
193. 1.9×10^{-35} moles
194. 4.77L
195. 8.75 mL
196. Single-Displacement
197. 3.27 g
198. $3\text{NH}_4\text{Br} + \text{FePO}_4 \rightarrow \text{FeBr}_3 + (\text{NH}_4)_3\text{PO}_4$
199. $0.0666 \frac{\text{mol}}{\text{L}}$
200. $3\text{C} + 4\text{Al}(\text{OH})_3 \rightarrow 6\text{H}_2\text{O} + 3\text{CO}_2 + 4\text{Al}$